Smaller explicit Diophantine forms of ZFC consistency

Twelve positive variables, twenty-four integers,
and a smaller fixed coefficient tuple

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Abstract

We combine and refine the constructions in the audited ProveIt snapshot `a477d0fc8b6f` (full pin in the references). A new equation in twelve positive integers has exact degree 1081 and a 116-operation arithmetic circuit; its three-equation version has maximum degree 1072 and 110 operations. A different construction uses twenty-four unrestricted integers, degree 2432, and 163 operations; its three-equation version uses 157 operations and has maximum degree 2416. A new parity layout reduces the largest fixed coefficient from 13,201,251 to 9,766,251 bits while retaining the 943-tile computation language. Ordinary mathematics formalizable in ZFC proves that each presentation is solvable exactly when ZFC has a finite refutation, by composition with the existing inputless compiler and exact machine calibration. The archive supplies literal circuits, exact compressed coefficient integers, finite recipes, proofs, independent reviews, and offline checks. These are ordinary proofs, not new Lean or Rocq formalizations. No global minimum or consistency proof is claimed.

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1 Results and the meaning of simplicity

We seek a fixed polynomial P with integer coefficients such that

$$\mathrm{ZFC} \vdash \left(\neg\exists \mathbf{x} \in D^n \; P(\mathbf{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}), \tag{1.1}$$

with an explicitly stated domain D. Here $\mathbb{N} = \{0, 1, 2, \ldots\}$; positive integers are $\mathbb{Z}_{>0}$. Consistency means the absence of a finite syntactic refutation in the recursive presentation used by the repository's exact machine endpoint.

There is no specified scalar measure of simplicity. We therefore keep four measures separate: number of unknowns, degree, arithmetic circuit length, and fixed coefficient size. A circuit is a literal finite polynomial description: every gate uses integer addition, subtraction, or multiplication. Shared expressions are abbreviations, not extra existential variables.

New construction	Domain	Unknowns	Equations	Degree	Operations
Relative root product	$\mathbb{Z}_{>0}$	12	1	1081	116
Relative root system, shifted	$\mathbb{Z}_{>0}$	12	3	1072	110
Signed saturation product	$\mathbb{Z}$	24	1	2432	163
Signed saturation system, shifted	$\mathbb{Z}$	24	3	2416	157
Signed saturation sum of squares	$\mathbb{Z}$	24	1	4832	166

Table 1: For a system, degree means maximum residual degree. Every binary operation is charged, including multiplication by a fixed numeral; comparisons are free under the repository convention.

The positive product improves the previously recorded best degree among the moderate-degree twelve-positive-variable equations, 1145 at 123 operations. It matches the best 116-operation count while reducing that construction's degree from 1353. The integer product improves the variable count, operation count, and degree of every listed wave-3 twenty-six-integer product. These are comparisons with the inspected repository family.

1.1 The inspected baseline

The audited reconciliation [1] records eight wave-3 packages, numbered 16–23. Earlier packages survive in history and in reference copies. The relevant recorded frontier is:

Package 18's exact degree is $161844 \cdot 5^{33} + 484$. It remains preferable when the sole objective is variable count in that domain. The 67-operation equation and the low-degree systems also remain useful. We found no justified improvement below 67 operations for a positive-domain single polynomial or below 59 for the smallest recorded arithmetic system.

Package 20 supplies the direct root merger and rational inverse formulas. Package 21 supplies an economical norm evaluator and Pell saturation. Package 22 supplies the transport shift. Package 23 excludes a negative odd scale after reconstructing the core. Package 19 supplies the Wang943 language. The new results combine these compatible components through one precise interface.

2 The inherited arithmetic and logical interface

2.1 Four fixed integer coefficients

The coefficient ports are

$$\beta = B - 1, \qquad \kappa = D_C + BD_R, \qquad M_C, \qquad \widehat{M}_F = M_F + B - 1, \qquad B = 2^{d_c}. \tag{2.1}$$

Recorded construction	Domain	Unknowns	Degree	Operations
Wave-1 P_{67}, retained in 17/21	$\mathbb{Z}_{>0}$	15	155	67
Package 22, retained system	$\mathbb{Z}_{>0}$	16	24	59
Package 20, lowest-degree root product	$\mathbb{Z}_{>0}$	12	1145	123
Package 21, shortest root product	$\mathbb{Z}_{>0}$	12	1353	116
Packages 21/22, root systems	$\mathbb{Z}_{>0}$	12	1344/1520	112
Package 19, lower-degree integer product	$\mathbb{Z}$	26	2465	192
Package 21, shorter integer product	$\mathbb{Z}$	26	2657	168
Package 23, low-degree integer product	$\mathbb{Z}$	27	172	116
Package 18, Jones specialization	$\mathbb{N}$	9	$\sim 1.9 \cdot 10^{28}$	20531

Table 2: Recorded baseline, not a fresh rebuild of every historical package. The new work directly inspects the relevant proofs and circuits in packages 19–23.

M_F is the native, unshifted high mask. The circuit port `MF` is $\widehat{M}_F$, and `K0` is κ. The four ports are fixed integers, not additional variables. They satisfy

$$\begin{gathered} 0 < M_C, M_F < B-1, \quad M_C \equiv 2 \pmod 4, \quad M_F \equiv 4 \pmod 8, \\ \operatorname{popcount}(M_C) + \operatorname{popcount}(M_F) = d_c. \end{gathered} \tag{2.2}$$

These conditions alone do not establish a compiler: the tile semantics, alignment, carry bounds, and canonical completeness are also needed. All four coefficients must be taken from one complete tuple. Section 7 specifies the new exact tuple.

2.2 Common polynomial abbreviations

The twelve core coordinates are

$$J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, r. \tag{2.3}$$

Their domains will differ. Define

$$\begin{aligned} &q = \beta J + 1, && C = q - F - \alpha, && X = wq, && Y = sq^3, && E = XY, \\ &k = \eta + \zeta, && c = kY + \eta, && a = E + Y, && H = 4a + 3, && \Delta = a^2 + H, \\ &\ell = EkY, && A = 1 + \ell(\ell + k), && G = 1 + \Delta c^2, && b_0 = X + ac. \end{aligned} \tag{2.4}$$

In particular $\Delta = (a+2)^2 - 1$. The rank expression and outer factors are

$$R = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + q\widehat{M}_F)J, \tag{2.5}$$

$$N_k = k - hE - R, \qquad N_t = (\kappa + w)C + (q - F) - z(q-1). \tag{2.6}$$

The strong and auxiliary quantities are

$$\begin{aligned} &I = ic^2, && D = 1 + \Delta I^2, && Q = \Delta(D-1) = (\Delta I)^2, && d = Q - 1, \\ &v = cT, && b = c + RD, && e = Q(Dv^2 + b^2) - 1, && B_a = 2Qvb. \end{aligned} \tag{2.7}$$

Every letter in these displays is a polynomial abbreviation. All exponents are fixed nonnegative integers.

2.3 The reused core theorem

The inputless proofs [3, 4, 5, 9] give an interface with reconstructed integer roots τ, u, f, y and integer γ:

$$\begin{aligned} \tau^2 - \ell(\ell+k) &= 1, \\ u = b_0 + H\gamma, \qquad u^2 - \Delta c^2 &= 1, \\ f^2 - \Delta i^2 c^4 &= 1, \\ Q[c(Tf-1) - Rf^2]^2 - (Q-1)y^2 &= 1. \end{aligned} \tag{2.8}$$

When $J, F, \alpha, z, s, w, \eta, \zeta > 0$ and $i \neq 0$, the other coordinates h, T, γ, f, τ, y may be signed. Together with $N_k = N_t \in \{1, -1\}$, these equations force both outer factors to $+1$ and decode a blank halt of K215. The ordinary-strong interface is obtained using $i_{\text{ordinary}} = \Delta|i|$, leaving Q unchanged.

The proof initially gets only $C \geq 0$, not $C > 0$. Its bounds are

$$q \geq 16, \quad q(q^2-1) < R < q^4 - 2, \quad R - 2 > 3q + 1. \tag{2.9}$$

With positive scales $E \geq q^4 > R + 2$. The source Pell-rank argument recovers the exact main rank R before settling the unit signs and extracting the tableau. This proof order preserves the exceptional $C = 0$ boundary correctly.

Conversely, a finite blank halt has canonical positive witnesses. Independent width/height padding by powers of five can simultaneously give odd i, square w, and odd s. The first two choices use the normalized strong index $m = cR$ and an odd exponent of q; the exact population calculation for s is recalled below. These choices preserve existence and are not claimed for every arbitrary earlier witness.

2.4 The ZFC endpoint and proof status

The inspected Lean endpoint [2] contains

$$\text{BlankHalts}(\mathsf{K215}) \longleftrightarrow \text{BProv}(\text{ZFCax}_s, \bot), \tag{2.10}$$

named `KEndpoint215.blank_halts_iff_zfc_refutation`. Its nonhalting theorem is obtained by negation. The ordinary finite-computation and syntactic-refutation calibration is the logical input here. We do not infer internal ZFC provability from an assertion that Lean is sound. The compiler and the new reductions are ordinary mathematics internalizable in ZFC; their composition proves (1.1).

No endpoint build was rerun, and the new machine-to-Diophantine composition is not a newly checked Lean or Rocq theorem. The finite audits certify their stated finite properties, while the mathematical arguments establish the unbounded equivalences.

3 A polynomial norm without hidden witnesses

Reduce $p(t)$ modulo the monic polynomial $t^2 - g$ to $p_0 + p_1 t$ and define

$$\text{Norm}_{g,t}(p) = p_0^2 - g p_1^2. \tag{3.1}$$

This is an explicit finite operation on polynomials. It adds no existential root and requires no root extraction at evaluation time. For polynomial weights L, M and $W = b_0 + Hr$, put

$$\begin{aligned} \mathcal{N}(L, M, W) = \text{Norm}_{G,u}\,\text{Norm}_{A,t}([d(W - Lu - HMt)^2 - H^2 e]^2 \\ - H^4 B_a^2 D). \end{aligned} \tag{3.2}$$

Equivalently this is the eight-factor product

$$\prod_{\substack{u=\pm\sqrt{G}\\ t=\pm\sqrt{A}\\ f=\pm\sqrt{D}}} (d(W-Lu-HMt)^2 - H^2e + H^2B_af). \tag{3.3}$$

All branches are counted with multiplicity, including coincident quadratic extensions.

3.1 A literal evaluator

The following expressions expand the norm using only $+, -, \times$:

$$\begin{aligned}
&Z_0 = W^2 + L^2G, && Z_1 = 2WL, && n = (HM)^2A,\\
&C_0 = d(Z_0 + n) - H^2e, && C_1 = dZ_1, && K_f = 4d^2n,\\
&K_0' = K_fZ_0, && K_1' = K_fZ_1, && j = H^4B_a^2D,\\
&V_0 = C_0^2 + GC_1^2 - K_0' - j, && V_1 = 2C_0C_1 - K_1',\\
&N_0 = V_0^2 + GV_1^2 - 4jK_0', && N_1 = 2V_0V_1 - 4jK_1',\\
&\mathcal{N} = N_0^2 - GN_1^2.
\end{aligned} \tag{3.4}$$

The scalar j is evaluated as $4H^4PU$, where $P = QDv^2$ and $U = Qb^2$. The identity

$$(C^2 + j - K)^2 - 4jC^2 = (C^2 - K - j)^2 - 4jK \tag{3.5}$$

in the quadratic ring explains the economical evaluator. K_0' is unrelated to the fixed coefficient κ.

Lemma 3.1 (Unit guard)**.** *For any integer assignment,* $8 \mid N_1$*, so* $\mathcal{N} \equiv N_0^2 \pmod{64}$*. Therefore*

$$N_kN_t(1+\mathcal{N}) = 1 \implies \mathcal{N} = 0, \quad N_k = N_t \in \{1, -1\}. \tag{3.6}$$

Proof. Z_1, C_1 are even, $2C_0C_1$ is divisible by four, and K_1' is divisible by eight. Hence $4 \mid V_1$ and $8 \mid N_1$. At a product zero all three factors are integer units. If $1 + \mathcal{N} = -1$, then $\mathcal{N} = -2 \equiv 2 \pmod 4$, contrary to its square residue. Thus the last factor is 1. □

4 Rational Pell saturation

Lemma 4.1. *If* $m \geq 2$ *and* V *are integers and* $y \in \mathbb{Q}$ *satisfies*

$$m^2V^2 - (m^2 - 1)y^2 = 1, \tag{4.1}$$

then y *is an integer.*

Proof. Write $y = A_0/B_0$ in lowest terms with $B_0 > 0$. Equation (4.1) gives $B_0^2 \mid m^2 - 1$. The positive integer $d_0 = (m^2 - 1)/B_0^2$ is nonsquare. The two positive integer-coordinate Pell solutions are (m, B_0) and $(m|V|, |A_0|)$; their ordinates are nonzero. Write the powers of the least positive integer-coordinate norm-one unit as (χ_j, ψ_j). For positive j, l,

$$m = \chi_j,\ B_0 = \psi_j,\ m|V| = \chi_l,\ |A_0| = \psi_l.$$

The recurrence gives $\chi_{n+2j} \equiv -\chi_n \pmod{\chi_j}$ and $\chi_{2j-n} \equiv -\chi_n \pmod{\chi_j}$. Reduce l modulo $2j$ and reflect to $0 \leq n \leq j$. Since $0 < \chi_n < \chi_j$ for $n < j$, divisibility $\chi_j \mid \chi_l$ forces l to be an odd multiple of j. Pell multiplication gives $\psi_j \mid \psi_l$, hence $B_0 \mid A_0$; coprimality forces $B_0 = 1$. □

This saturation lemma is inherited from packages 19–22 [7, 6, 8]. It closes an integrality gap that rationality alone would not close for a general nonmonic quadratic. The new work combines it with shorter weights and with a signed scale that removes two positive-domain witnesses.

5 A smaller equation in twelve positive integers

Use all twelve coordinates in (2.3) as positive integers. Define

$$M = kI^2 + T^2, \qquad L = 1 + 2HM, \qquad W = b_0 + Hr. \tag{5.1}$$

The proposed single equation is

$$\boxed{P_{12} := N_k N_t(1 + \mathcal{N}(L, M, W)) - 1 = 0.} \tag{5.2}$$

Together with the finite definitions in Sections 2 and 3, this is a fixed integer polynomial, not an exponential Diophantine equation.

Theorem 5.1. *For the complete compiler tuple of Section 7, $P_{12} = 0$ has a solution in twelve positive integers if and only if* K215 *halts on its blank tape. Its exact total degree is* 1081*; the supplied circuit uses* 116 *binary arithmetic operations.*

5.1 Weak bounds before root recovery

By Lemma 3.1, a product zero gives $\mathcal{N} = 0$ and $N_k = N_t = \epsilon \in \{1, -1\}$. The inherited transport bootstrap is unchanged, since it involves only the outer expressions. It gives

$$\begin{gathered} q \geq 16, \quad X \geq 16, \quad Y \geq 4096, \quad C \geq 0, \quad 0 < R < q^4 - 2, \quad R > 3q \geq 48, \\ E \geq q^4 > R + 2, \qquad k = R + \epsilon + hE > E, \qquad R < k, \\ kY < c < k(Y + 1) < 2kY. \end{gathered} \tag{5.3}$$

No square-root recovery, sign of γ, or strict inequality $C > 0$ is assumed in this step.

5.2 The inner weight

Write $a_0 = a + 2$ temporarily. Since $X \geq 16$, the weak bounds give $a_0 < 2E$. They also give

$$\ell = kEY > \frac{a_0 c}{4}, \qquad \ell > E^2 Y > 8a_0^2,$$

and therefore

$$A > \frac{a_0^2 c^2}{16}, \qquad A > 64a_0^4. \tag{5.4}$$

As $I = ic^2$ with $i \geq 1$, we have

$$D < a_0^2 I^2, \qquad c < I^2, \qquad b = c + RD < 2ka_0^2 I^2.$$

Using $M^2 \geq 4kI^2T^2$ and $k > 48$, we obtain two separate estimates:

$$AM^2 > \frac{k}{4} Dv^2 > 8Dv^2, \qquad AM^2 \geq Ak^2 I^4 > 16b^2 > 8b^2.$$

Averaging them, and using $d = Q - 1 \geq Q/2$, yields

$$\boxed{dAM^2 > 2Q(Dv^2 + b^2) = 2(e + 1).} \tag{5.5}$$

This strengthening of the inherited constants allows us to use $M = kI^2 + T^2$ without an extra factor of two.

For either root $f' = \pm\sqrt{D}$, any complex y' satisfying $d(y')^2 = e - B_a f'$ obeys

$$\begin{aligned} d|y'|^2 &\le e + B_a\sqrt{D} < Q(v\sqrt{D} + b)^2 \\ &\le 2Q(Dv^2 + b^2) < dAM^2. \end{aligned}$$

Thus every such branch satisfies

$$|y'| < M\sqrt{A}. \tag{5.6}$$

The complex absolute value matters: the right-hand side of the quadratic equation for a particular branch can be negative.

5.3 Relative separation of the two main-root branches

The decisive simplification is the comparison $A < G$:

$$\begin{aligned} G - A &> k^2\big((E+Y)^2Y^2 - EY(EY+1)\big) \\ &= k^2\big(EY(2Y^2-1) + Y^4\big) > 0. \end{aligned} \tag{5.7}$$

Let $\Phi(x) \in \mathbb{Z}[x]$ be the product over $t = \pm\sqrt{A},\ f = \pm\sqrt{D}$ of

$$d(x - Mt)^2 - e + B_a f.$$

Every complex root of Φ has the form $x = Mt + y'$, so (5.6) gives the uniform bound

$$|x| < 2M\sqrt{A} < 2M\sqrt{G}. \tag{5.8}$$

At $\mathcal{N} = 0$, one root $u = \pm\sqrt{G}$ gives $x = (W - Lu)/H$ with $\Phi(x) = 0$. If G is not a square, conjugation in $\mathbb{Q}(\sqrt{G})$ also makes $x' = (W + Lu)/H$ a root. But

$$|x' - x| = \frac{2L\sqrt{G}}{H} > 4M\sqrt{G},$$

contradicting the sum of the two bounds (5.8). Consequently G is a square and u is an integer. Moreover $u > 0$: if $u < 0$, then

$$W = Lu + Hx < -L\sqrt{G} + 2HM\sqrt{G} = -\sqrt{G} < 0,$$

whereas $W = b_0 + Hr > 0$. We have recovered an integer main root and a rational x.

The earlier absolute separation used an additional factor $\ell + k$ in L. Comparing the inner radius with the distance between the two main-root branches removes that factor.

5.4 Recovering the remaining integer roots

For this rational x, put

$$\mathcal{C} = d(x^2 + M^2A) - e, \qquad \mathcal{V} = \mathcal{C}^2 + DB_a^2 - 4d^2x^2M^2A. \tag{5.9}$$

Equation (5.5) implies $\mathcal{C} > e + 2 > 0$. Also $x \neq 0$, since

$$\Phi(0) = \big[(dM^2A - e)^2 - DB_a^2\big]^2 > 0,$$

using $B_a\sqrt{D} \le e + 1$. The coherent inverse formulas inherited from the direct merger are

$$f = -\frac{\mathcal{V}}{2\mathcal{C}B_a}, \qquad t = \frac{\mathcal{C} + B_a f}{2dxM}, \qquad y = x - Mt. \tag{5.10}$$

Their denominators are nonzero. Squaring the original factor and eliminating $t^2 = A$ gives $\mathcal{V}+2\mathcal{C}B_a f = 0$; substituting back gives the displayed expression for t. Thus (5.10) supplies rational numbers with $f^2 = D$, $t^2 = A$, and

$$Q(vf - b)^2 - (Q-1)y^2 = 1. \tag{5.11}$$

Rational square roots of integers are integers, so $f, t \in \mathbb{Z}$. Since $Q = (\Delta I)^2$ and $vf - b \in \mathbb{Z}$, Lemma 4.1 implies $y \in \mathbb{Z}$.

Now $x = Mt + y$ is integral. Reducing $W = Lu + Hx$ modulo H, and using $L \equiv 1 \pmod{H}$, gives

$$\gamma = \frac{u - b_0}{H} \in \mathbb{Z}.$$

In fact $\gamma > 0$: from $c > k > E > X$, $\Delta > (a+1)^2$, and $u > 0$, we get $u > (a+1)c$, hence $\gamma H > c - X > 0$. The identity

$$vf - b = c(Tf - 1) - Rf^2$$

uses $f^2 = D$ and restores all four equations (2.8). Their permitted signed quotient absorbs a negative f by changing (f, T) to $(-f, -T)$. The inherited core theorem forces $\epsilon = 1$ and decodes a blank halt of K215.

Conversely, take a canonical positive normalized compiler witness. Choose its positive roots u, t, y and set

$$r = \gamma + 2Mu + Mt + y > 0. \tag{5.12}$$

Then $W = Lu + H(Mt + y)$, so one factor in (3.3) vanishes. The outer factors both equal one. This proves both directions of Theorem 5.1.

5.5 A three-equation version in 110 operations

Initially the equivalent system is

$$N_k = 1, \qquad N_t = 1, \qquad N_0^2 = GN_1^2. \tag{5.13}$$

Direct comparison in the last equation saves the subtraction defining $\mathcal{N}$. For the shorter system replace the old transport coordinate by $z_{\text{old}} = z + 1$ and use

$$\boxed{\begin{aligned} N_k &= 1, \\ (\kappa + w)C &= F + z(q-1), \\ N_0^2 &= GN_1^2. \end{aligned}} \tag{5.14}$$

Every positive solution of (5.14) gives one of (5.13). Conversely, after the compiler has been recovered, it gives $C > 0$ and $X = wq = 2^R$ with $R > 3q$. Hence $w > q > F$, and

$$(\kappa + w)C - F = (z_{\text{old}} - 1)(q - 1) > 0.$$

Thus $z_{\text{old}} \geq 2$, making $z = z_{\text{old}} - 1$ positive. This is the zero-set shift from package 22 [8]; it is used after soundness, not as a premature weak bound. The product (5.2) retains the old coordinate.

The product ledger is 74 multiplications and 42 additions or subtractions. The shifted system uses $72 + 38 = 110$ operations; its residual degrees are $7, 2, 1072$.

5.6 Exact degree certificate

Give all twelve supplied variables degree one and the four ports degree zero. The relevant degrees are

	c	H	Δ	I	D	Q	R	b	e
deg	5	6	12	11	34	46	4	38	122

and

	A, G	M	L	W	n	Z_0	Z_1
deg	22	23	29	11	80	80	40

The second array has a shared A, G entry. In C_0 the unique highest homogeneous part is $-H^2 e$, of degree 134, while $d(Z_0 + n)$ has degree at most 126. It follows successively that V_0 has highest part $H^4 e^2$, of degree 268, and N_0 has highest part $H^8 e^4$, of degree 536. In contrast, $\deg N_1 \le 488$. Therefore the final terms have degrees 1072 and at most 998; their leading terms cannot cancel.

More explicitly, with $q_0 = \beta J,\ k_0 = \eta + \zeta$,

$$\mathcal{N}_{\mathrm{top}} = 2^{32} i^{48} k_0^{96} w^{80} s^{176} q_0^{656} (q_0 - F)^{16}. \tag{5.15}$$

The product's highest part is this times

$$(-hwsq_0^4)\,[w(q_0 - F - \alpha) - zq_0].$$

The coefficient of

$$J^{677} hz i^{48} \eta^{96} w^{81} s^{177}$$

is $2^{32}\beta^{677} \neq 0$. Thus $\deg \mathcal{N} = 1072$ and $\deg P_{12} = 1081$ for every authentic tuple, not just a generic small specialization.

6 Twenty-four unrestricted integer unknowns

The positive proof above does not extend merely by allowing s to be negative: its comparison $A < G$ and the signs of its weights were proved on the positive domain. For the integer construction we instead retain the polynomial norm circuit of package 21 and prove a larger domain for it.

Use the common abbreviations (2.4)–(2.7), but now set

$$\mathcal{B} = v^2 + D + 2b, \qquad L = 1 + H\mathcal{B}, \qquad M = L, \qquad W = b_0 + Hr. \tag{6.1}$$

Require positivity only for

$$J, F, \alpha, z, \eta, \zeta.$$

For each of these six coordinates use a distinct triple of integer unknowns and the polynomial

$$\mathrm{Pos}(x, y, t) = 1 + x^2 + y^2 + t(t + 1). \tag{6.2}$$

It is positive for every integer triple and maps onto the positive integers. Indeed $4n - 3 \equiv 1 \pmod 4$ is a sum of three squares with one odd square and two even squares; dividing after separating the odd square gives (6.2).

For the remaining coordinates put

$$i = 2i_0 + 1, \qquad s = 2s_0 + 1, \qquad w = w_0^2, \qquad h, T, r \in \mathbb{Z}. \tag{6.3}$$

There are $6 \cdot 3 + 6 = 24$ independent integer unknowns. After these substitutions the polynomial is

$$\boxed{P_{24} := N_k N_t\big(1 + \mathcal{N}(L, L, W)\big) - 1 = 0.} \tag{6.4}$$

Theorem 6.1. *For the complete compiler tuple of Section 7, $P_{24} = 0$ has an integer solution if and only if* K215 *halts on its blank tape. Its exact degree is* 2432*, and its circuit uses* 163 *binary operations.*

6.1 Bounds valid for both signs of the scale

The unit guard again gives $\mathcal{N} = 0$ and $N_k = N_t = \epsilon \in \{1, -1\}$. Positivity of F, z implies

$$(q - F) - z(q - 1) = 1 - F - (z - 1)(q - 1) \le 0.$$

Since $\kappa + w \ge 2$, the alternative $C \le -1$ would make $N_t \le -2$. Hence $C \ge 0$, and the unchanged mask argument gives the bounds (2.9).

Put $Z = |Y|$. The odd substitution excludes $s = 0$, so

$$\begin{gathered} Z \ge q^3, \quad X \ge 0, \quad |a| = (X + 1)Z, \quad k(Z - 1) < |c| < 2kZ, \quad |H| > 1, \\ \Delta \ge (q^3 - 1)(q^3 - 3) > q^4 > R. \end{gathered} \tag{6.5}$$

Since $i \ne 0$ and $|c| > 1$,

$$D > \Delta + |c|, \qquad 1 < b = c + RD < Q. \tag{6.6}$$

For the upper bound, explicitly,

$$Q - b = (\Delta - R)D - \Delta - c \ge D - \Delta - |c| > 0.$$

Also $G, D, Q > 1$ and $\ell = XkY^2 \ge 0$, so $A \ge 1$. In particular $\mathcal{B} > 0$, even with signed or zero T.

6.2 Root recovery with possibly negative weights

The factorization of the norm supplies a coherent triple $u^2 = G,\ t^2 = A,\ f^2 = D$ and

$$y = \frac{W - Lu - HLt}{H}, \qquad dy^2 = e - B_a f = Q(vf - b)^2 - 1. \tag{6.7}$$

Work in a finite normal splitting field. Every conjugate y' satisfies $|y'| < \mathcal{B}$. To see this, put $M_0 = |v|\sqrt{D} + b > 1$. The real quantity $Q(vf' - b)^2 - 1$ lies between -1 and $QM_0^2 - 1$, so its absolute value is at most $QM_0^2 - 1$. Thus

$$|y'|^2 \le \frac{QM_0^2 - 1}{Q - 1} < 2M_0^2, \qquad \mathcal{B} = v^2 + D + 2b \ge 2M_0. \tag{6.8}$$

This bounds imaginary as well as real branches.

Set $U_0 = |H|\mathcal{B} > 2$. Although L can be negative, $|L| \ge U_0 - 1$. Since $\sqrt{G} > 2$, we obtain

$$|L|\sqrt{G} > U_0. \tag{6.9}$$

If $X > 0$, then $X \ge q$, and

$$\sqrt{A} > \ell = kZ|E|, \qquad \sqrt{G} < (|a| + 2)|c| < 2(|a| + 2)kZ.$$

Consequently

$$\begin{aligned} |H|\sqrt{A} - \sqrt{G} &> kZ((4|a| - 3)|E| - 2(|a| + 2)) \\ &\ge kZ(2|a| - 7) > 2. \end{aligned}$$

Using $2|L| > U_0$, this implies the second strict separation

$$|HL|\sqrt{A} > |L|\sqrt{G} + |H|\mathcal{B}. \tag{6.10}$$

If t were irrational, an automorphism of the splitting field would flip its sign. Subtracting the conjugates of $W = Lu + HLt + Hy$ would contradict (6.10), because the other two differences have absolute values at most $2|L|\sqrt{G}$ and strictly less than $2|H|\mathcal{B}$. Thus t is rational. If $X = 0$, then $A = 1$, so t is rational directly and (6.10) is unnecessary.

With t rational, an automorphism flipping u would contradict (6.9). Hence u is rational as well. Both are integers because their squares are integers; equation (6.7) then makes y rational.

6.3 Saturation and the two exceptional boundaries

If $T \neq 0$, then $v \neq 0$, $B_a \neq 0$, and

$$f = \frac{e - dy^2}{B_a} \in \mathbb{Q}.$$

Since $f^2 = D$, this root is integral. Now $vf - b \in \mathbb{Z}$, so Lemma 4.1 applied with $m = |\Delta i c^2|$ makes y integral. If $T = 0$, the same lemma applies directly with $V = b$, without first recovering f, and again gives $y \in \mathbb{Z}$. Because $L \equiv 1 \pmod{H}$, in either case

$$\gamma = \frac{u - b_0}{H} = r - \mathcal{B}u - Lt - y \in \mathbb{Z}. \tag{6.11}$$

If $X = 0$, then $u = ac + H\gamma$. Reducing $u^2 - \Delta c^2 = 1$ modulo H, with $\Delta = a^2 + H$, gives $H \mid 1$, contrary to $|H| > 1$. Therefore $X > 0$ and $w > 0$. If $T = 0$, (6.7) gives

$$y^2 = b^2 + \frac{b^2 - 1}{Q - 1}.$$

The integer inequalities $1 < b < Q$ put this number strictly between b^2 and $(b+1)^2$, a contradiction. Thus $T \neq 0$, and f is integral in all possible cases.

We have now reconstructed all four equations (2.8), with signed t, u, f, y, γ, h, T. No division by an unproved nonzero T or B_a was used.

6.4 Why a negative odd scale is impossible

The recovered equations meet the hypotheses of the signed-scale exclusion in package 23 [9]. The normalized multiplier need only be nonzero: replacing i by $|i|$ changes none of the equations. For clarity, the mechanism of that exclusion is as follows. If $s < 0$, put

$$Z = -Y, \qquad B_0 = (X+1)Z - 2, \qquad c_0 = -c, \qquad P_0 = 2XZ^2 + 1.$$

The first and main Pell classifications give indices n, p with $k = 2\psi_{P_0}(n)$ and $c_0 = \psi_{B_0}(p)$. Here ψ_B denotes the ordinate of $(B + \sqrt{B^2 - 1})^j$. The normalized strong equation makes its index divisible by pc_0. The auxiliary congruences then recover $p = R$; the sign of c disappears in the squared congruence. The strict index interval forces $\epsilon = 1$ and $R = 2r_0 + 1$, and the Pell growth comparison gives

$$Z > \frac{X^{r_0}}{2} > 2^R.$$

The two possible signs of the main root give respectively

$$X + 2^R \equiv 0 \pmod{H} \quad \text{or} \quad X\,2^R \equiv 1 \pmod{H}.$$

Both are excluded by

$$0 < X + 2^R < |H|, \qquad 1 < X\,2^R < XZ < |H|.$$

The full inherited rank argument, including its signed congruences, is supplied in the reference copy of `ODD_SCALE_27_PROOF.md`. It applies to the reconstructed core, not to a merely rational norm witness. This order is essential.

Since s is odd and cannot be negative, it is positive. The signed normalized compiler now decodes a blank `K215` halt and forces $N_k = N_t = 1$.

6.5 Simultaneous completeness of the domain restrictions

From an actual blank halt, canonical independent width and height padding gives $q = 2^{\theta}$ with $\theta = d_c N$ odd. The canonical rank R is odd, so

$$w = 2^{R-\theta}$$

is a positive square. Choosing the normalized strong index $m = cR$ gives odd positive i, together with the required auxiliary quotient, as in the signed-domain proof [5]. These restrictions concern the same padded tableau.

To check odd s without changing that witness, set

$$S = (C-1) + qF, \qquad T_{\mathrm{mask}} = (M_C J + 1) + q(M_F J - 1), \tag{6.12}$$

where M_F is the *native* high mask, not the shifted circuit port $\widehat{M_F}$. The native mask constraints give $S \,\&\, T_{\mathrm{mask}} = 0$, both quantities below q^2, and $\mathrm{popcount}(T_{\mathrm{mask}}) = \theta + 2$. The exact packing identity is

$$R = (q^2 - 1 - S)q^2 + (S + T_{\mathrm{mask}}).$$

The two blocks are disjoint, so

$$\mathrm{popcount}(R) = 2\theta + \mathrm{popcount}(T_{\mathrm{mask}}) = 3\theta + 2.$$

The inherited half-binomial valuation gives $v_2(Y) = \mathrm{popcount}(R) - 2 = 3\theta$. Therefore $s = Y/q^3$ is positive odd. Every remaining guarded coordinate is in the image of (6.2). Finally, with t again the canonical first Pell root, choose

$$r = \gamma + \mathcal{B}u + Lt + y. \tag{6.13}$$

This integer makes one norm factor vanish, proving completeness for (6.4) with all three restrictions satisfied simultaneously.

6.6 The system and sum-of-squares alternatives

The unshifted system is $N_k = 1,\ N_t = 1,\ \mathcal{N} = 0$. For the recommended shorter system, use six positive maps including the new z, put $z_{\mathrm{old}} = z + 1$, and write

$$\boxed{\begin{aligned} N_k &= 1, \\ (\kappa + w)C &= F + z(q-1), \\ N_0^2 &= G N_1^2. \end{aligned}} \tag{6.14}$$

The proof of the shift is the same as for (5.14), after the core has been decoded. This system has 24 integer unknowns, three equations, 157 gates, and residual degrees $12, 4, 2416$. The unshifted version uses 159 gates. For a single polynomial obtained directly from the system, the archive also includes

$$(N_k - 1)^2 + (N_t - 1)^2 + \mathcal{N}^2 = 0, \tag{6.15}$$

with degree 4832 and 166 operations. Its integer zero set is exactly that of the unshifted system.

6.7 Operation count and exact degree

Each positive map costs seven operations, the two odd maps cost two each, and the square map costs one. The total domain-conversion cost is therefore $42 + 4 + 1 = 47$. The 116-gate package-21 product circuit is retained literally after that prefix, yielding 163 gates. This is a new theorem about the domain of an old polynomial circuit, as well as a smaller explicit integer presentation.

Give $J, F, \alpha, z, w, \eta, \zeta$ weight two and the other core coordinates weight one. Key weighted degrees are

	c	H	Δ	A, G	D	Q	$b, \mathcal{B}$	L
deg	9	11	22	40	60	82	68	79

In the evaluator (3.4), n has weight 220, and the unique highest part of C_0 is dn, of weight 302. The highest part of N_0 has weight 1208, while $\deg N_1 \le 1087$. Thus N_0^2 has weight 2416 and GN_1^2 has weight at most 2214; there is no leading cancellation. With $q_0 = \beta J,\ k_0 = \eta + \zeta$, the highest weighted part of the norm is

$$2^{80} i^{48} k_0^{112} w^{112} s^{224} q_0^{832} (q_0 - F)^{16}. \tag{6.16}$$

Multiplying by the leading outer factors gives

$$\begin{aligned}(P_{24})_{\mathrm{top}} = \ & -2^{80} h i^{48} k_0^{112} w^{113} s^{225} q_0^{836} (q_0 - F)^{16} \\ & \cdot [w(q_0 - F - \alpha) - z q_0],\end{aligned} \tag{6.17}$$

before the actual degree-two substitutions. For a guarded coordinate p, denote the first variable in its positive-map triple by p_x. After substituting, the coefficient of

$$J_x^{1706} h z_x^2 i_0^{48} \eta_x^{224} w_0^{226} s_0^{225}$$

is $2^{353}\beta^{853} \neq 0$. Its degree is 2432. The analogous norm coefficient is $2^{352}\beta^{848} \neq 0$, at degree 2416. Consequently the product and system degrees hold for the actual fixed coefficients, uniformly in every authentic $\beta > 0$.

7 A smaller complete coefficient tuple

The preceding reductions can use a complete authentic tuple from the inherited compiler. We additionally give a smaller tuple for the same 943-tile Wang language. The improvement changes the arithmetic placement of tile-copy tests, while retaining the ordered language, the blank-machine semantics, and the radix-cell size.

7.1 Exact finite specification

The archive includes the ordered list of 943 tiles from package 19 [10], generated from the literal `K215` transition table. A tile is (x_W, x_N, x_E, x_S), with all labels below 512. Selector 0 is Start; selector 1 is the globally impossible Forbidden tile. The list's SHA-256 is

`66b55f1d11fca9fefc93b12fcfc8b4cabb75b`
`08e9fd03311f9e894a1c5e5e81f`.

The split display denotes one 64-digit hash with no separator. This fixed finite list, rather than a hypothetical halting run, is the input to the following recipe.

Set

$$A_c = 1024, \quad \rho = 2^{625}, \quad L_c = 15625, \quad d_c = 625 L_c = 9765625 = 5^{10}, \quad B = \rho^{L_c}. \tag{7.1}$$

For $j = 0, \dots, 9$, place the four kinds of native copy bits at

t	W	N	E	S
$e_t(j)$	$1020 + 2j$	$1000 + 2j$	$980 + 2j$	$960 + 2j.$

(7.2)

Let $\mathcal{E}$ be the disjoint union of selectors $0, \dots, 942$, these forty copy positions, dummy positions $1063, 1064$, and anchors $1095, 1096, 2195, 3296$. Thus $|\mathcal{E}| = 989$, and $\max \mathcal{E} = 3296$. Order directions as W, N, E, S, numbered $t = 0, 1, 2, 3$. For selector i, define the fixed integer

$$c_i = 1 + \sum_{t=0}^{3} \sum_{j=0}^{9} \mathrm{bit}_j(x_t + 512) A_c^{1+10t+j} + \sum_{j=41}^{44} A_c^j. \tag{7.3}$$

The remaining coefficients are

$$\begin{aligned} &c_{e_t(j)} = A_c^{1+10t+j}, \qquad c_{1095} = A_c^{41}, \quad c_{1096} = A_c^{42}, \\ &c_{2195} = A_c^{43}, \quad c_{3296} = A_c^{44}, \qquad c_{1063} = c_{1064} = 0. \end{aligned}$$

Put

$$\mu = A_c - 2 + \sum_{j=1}^{44} (A_c - 3) A_c^j + 2^{450}(2^{78} - 1).$$

Then the new directional coefficients are

$$\begin{aligned} D_C &= \rho^{40} + \rho^{41} + \rho^{1100} + \rho^{2200} \\ &\quad + \sum_{e \in \mathcal{E}} c_e (\rho^{7631-e} + \rho^{8733-e}), \\ D_R &= \rho. \end{aligned} \tag{7.4}$$

The native masks are

$$\begin{aligned} M_F &= 4 + \mu(\rho^{7631} + \rho^{8733}) + \sum_{j=0}^{9} \rho^{1000+2j} + \sum_{j=0}^{9} \rho^{1021+2j} + \rho^{2195} + \rho^{3296}, \\ M_C &= B - 1 - \sum_{\substack{e \in \mathcal{E} \\ e \neq 1}} \rho^e - 2\rho^{1063}. \end{aligned} \tag{7.5}$$

Finally take the four ports exactly as in (2.1). The code calls them `beta`, `K0`, `MC`, and `MF`; the stored payload for the last is `MF_source`. The archive contains all four integers as deterministic compressed binary payloads, their hashes, the transition table, the tile list, and independent reconstruction code. No variable exponent occurs in this definition: every sum is finite over a displayed fixed range or the supplied finite tile list.

7.2 Parity separates temporal and spatial tests

All native copy positions are even. The temporal copy tests are at the even positions $1000 + 2j$, while the spatial copy tests are at the odd positions $1021 + 2j$. This makes the native transport displacement 1 sufficient. The factor BD_R in κ still moves to a neighboring whole cell; only its displacement *within* the cell is reduced.

At temporal target $1000 + 2j$, the bare D_C-shift 40 sees exactly the south copy $960 + 2j$. The shift 41 and the D_R-shift 1 would see unsupported odd positions. The rotated native word contributes the north copy $1000 + 2j$. At spatial target $1021 + 2j$, the D_C-shift 41 sees the east copy $980 + 2j$, and

D_R's shift 1 sees the west copy $1020 + 2j$. The shift 40 and the rotated native word would require unsupported odd positions. Selectors end at 942, and the odd dummy and anchors are outside these tested ranges.

At anchor targets $2195, 3296$, the only bare-shift contributions are $1095 + 1100$ and $1096 + 2200$, respectively. The other bare shifts and D_R miss the native support. Every reflected clause exponent is at least $7631 - 3296 = 4335$, above all these targets. The twenty edge tests and two synchronization tests therefore have exactly their inherited meanings.

7.3 Clause isolation and alignment for arbitrary rotations

Write $\mathcal{E} = \mathcal{S} \cup \{2195, 3296\}$, with $\max \mathcal{S} = 1096$. The two clause bands differ by 1102. No pair in $\mathcal{S}$ has that difference, and

$$2195 - 1102 = 1093 \notin \mathcal{E}, \qquad 3296 - 1102 = 2194 \notin \mathcal{E}.$$

Since $3296 - 2195 = 1101$, we have $1102 \notin \mathcal{E} - \mathcal{E}$. At either clause target the bare shifts and D_R also miss $\mathcal{E}$. Thus its coefficient is exactly $\sum_e c_e d_e$, with the same selector and copy equations as the original language.

The upper-dummy spill position is $1063 + 1 = 1064$, already in $\mathcal{E}$. Moreover $L_c > 2 \max \mathcal{E}$. Hence the two circular sets

$$\{7631 - e \bmod L_c : e \in \mathcal{E}\}, \qquad \{8733 - e \bmod L_c : e \in \mathcal{E}\}$$

are disjoint. For every rotation, at least one clause band is clean.

For binary alignment, write a rotation as $625t + \varepsilon$, where $0 \le \varepsilon < 625$. The inherited carry-free bound implies that for $1 \le \varepsilon \le 623$ every rotated radix digit is even. For $\varepsilon = 624$, only the single upper-dummy spill can give an odd digit. At an occupied marked cell, both tested anchors require odd digits, excluding both kinds of misalignment. For $\varepsilon = 0$, the sole ordered support pair at difference 1101 is $(2195, 3296)$: $2195 - 1101 = 1094$ is absent, whereas $3296 - 1101 = 2195$ is present. There are no additional circular cases since $L_c > 2 \max \mathcal{E}$. The anchor tests therefore force whole-cell alignment. This checks arbitrary purported rotations; it does not assume a desired stride divisibility in advance.

7.4 Carry, population, origin, and canonical padding

The values c_e are only relocated. Their mass and the base-A_c clause inequalities remain unchanged. For completeness the finite bounds checked in the archive are

$$\begin{aligned} \text{mass} &= (989 + 2)\left(2\sum_e c_e + 6\right), \\ \text{threshold} &= \max(4\text{mass} + 8,\ 2\mu + 4,\ 16) < \rho, \\ \frac{3\rho}{4} + \text{mass} &\le \rho - 2, \qquad \frac{\rho}{2} + 1 + \text{mass} \le \rho - 2. \end{aligned} \tag{7.6}$$

The mass has 461 bits and the threshold 529 bits, versus 625 bits of radix capacity. The largest possible product exponents are bounded by

$$8733 + 3296 = 12029 < L_c, \qquad 1 + 3296 = 3297 < L_c.$$

There is no new intercell spill. The masks satisfy

$$\text{popcount}(M_F) = 989, \quad \text{popcount}(M_C) = d_c - 989, \quad M_C \equiv 2 \pmod 4, \quad M_F \equiv 4 \pmod 8,$$

and both native masks lie strictly between 0 and $B-1$. The low mask has its highest in-cell bit set, but neither mask has a bit at or above the cell cutoff.

The minimum exponents of D_C, D_R are $40, 1$, both positive. The low-origin unit therefore still comes only from the temporal neighbor's selector 0. The canonical unique Start is not its own temporal neighbor, preserving the special origin test.

Finally $\rho \equiv 2 \pmod 5$. The paired clause contributions cancel modulo five, because their exponent gap 1102 is 2 (mod 4). The bare terms give

$$D_C \equiv 1+2+1+1 = 0 \pmod 5, \qquad D_R \equiv 2 \pmod 5, \qquad 2D_C - D_R \equiv 3 \pmod 5. \tag{7.7}$$

Under the canonical power-of-five padding, let w_{tile} denote the tableau width (distinct from the core coordinate w). The dummy coefficient controlling the actual packed rank is

$$\Gamma = \rho^{1063}(q^2-1)\big[1+q(D_C + BD_R + B^{w_{\text{tile}}})\big] \equiv 2 \pmod 5. \tag{7.8}$$

It is a unit modulo the required power of five. Thus the inherited Boolean subset adjustment still realizes the actual-rank stride congruence. The unchanged tile language supplies independently large power-of-five width and height padding. Together with the preserved origin and strict slack bounds, this establishes canonical completeness. The mask calculation (6.12) is unchanged, so the simultaneous odd-i, square-w, odd-s refinements also survive.

Proposition 7.1. *The complete tuple (7.1)–(7.5) meets the native compiler's soundness and canonical completeness interface. Consequently it can be used in both Theorems 5.1 and 6.1.*

Proof. The edge, clause, occupancy, alignment, and carry arguments above decode exactly the same Wang tilings as the inherited 943-tile language. Its Start/Forbidden and helical computation theorem [10] decodes precisely blank halting computations. Conversely the same language pads an actual finite computation; (7.6)–(7.8) give the required packing, rank adjustment, and slack. The native arithmetic compiler proof [3, 4] therefore applies to the complete new tuple. The supplied proof notes spell out the same adapter independently of this compressed presentation. □

7.5 Exact size reduction

Since $0 < D_C < B$ and $D_R = 2^{625}$,

$$\text{bitlength}(\kappa) = d_c + 625 + 1 = 9{,}766{,}251.$$

This is the largest of the four actual port bit lengths.

Measure	Previous Wang943	New parity tuple	Reduction
Largest port, bits	13,201,251	9,766,251	3,435,000
Four raw payloads, bytes	5,312,269	4,882,894	429,375
Four deterministic gzip files, bytes	20,260	19,820	440

Table 3: The coefficient-height improvement is 26.02%. Compression lengths are separate storage measurements.

This adapter alone does not reduce the tile count, variable count, degree, or arithmetic operation count. It supplies a smaller exact numeric backend for both new arithmetic constructions.

8 Consistency equivalence, reproducibility, and limits

Theorem 8.1 (The requested equivalence)**.** *Let the fixed four integers be the tuple in Section 7. ZFC proves*

$$\begin{aligned} \neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{12}\, P_{12}(\mathbf{x}) = 0 \;&\longleftrightarrow\; \mathrm{Con(ZFC)}, \\ \neg\exists \mathbf{z} \in \mathbb{Z}^{24}\, P_{24}(\mathbf{z}) = 0 \;&\longleftrightarrow\; \mathrm{Con(ZFC)}. \end{aligned} \tag{8.1}$$

The same equivalences hold for the shifted three-equation systems (5.14), (6.14), in their stated domains, and for the integer sum of squares (6.15).

Proof. Theorems 5.1 and 6.1, with the coefficient adapter, give finite polynomial solutions if and only if there is a finite blank halting computation of the literal machine. The exact machine calibration identifies those computations with finite ZFC refutations. All reductions use finite coding, integer arithmetic, Pell recurrences, finite algebraic extensions, and the indicated classical number-theoretic facts; their proofs can be carried out in ZFC. Thus ZFC proves each existential statement equivalent to its syntactic refutability statement. Negating gives (8.1). The system shifts preserve existence in both directions, and over the integers a sum of squares vanishes exactly when every summand vanishes. □

This theorem does not decide which side holds. In particular, under the usual consistency assumption, proving one of these polynomials has no solutions within ZFC would contradict the second incompleteness theorem. The construction identifies the arithmetic statement whose unprovability is at issue.

8.1 What is literal and what is inherited

The archive is intended as a reviewable repository contribution. It contains:

- this LaTeX source and its PDF, with the new proofs and exact nonvanishing degree certificates;
- seven literal JSON arithmetic circuits: three for the positive construction and four for the integer construction;
- Python generators and finite arithmetic diagnostics, plus an independent circuit/liveness/degree-bound checker;
- the complete coefficient recipe, transition table, ordered Wang tile list, four compressed binary integers, and independent byte-for-byte reconstruction checks;
- independent written reviews and the source copies needed to audit the reused norm, signed-domain, compiler, and endpoint interfaces;
- source provenance, licenses, an artifact checksum manifest, and a root reproduction command.

The circuit files enumerate the supplied variables and the four fixed ports separately, define every intermediate using a binary operation, and identify their output equation or comparisons. They can be expanded mechanically to ordinary polynomials, although doing so loses useful sharing and greatly enlarges the presentation. The degrees in Table 1 are the degrees of those expanded polynomials, not circuit-depth estimates.

The newly supplied proofs establish the two root reductions and the coefficient adapter. The inherited native compiler, Pell-index lemmas, Wang language theorem, and machine/refutation calibration remain dependencies. Their pinned sources are identified and relevant copies are included. We do not claim to have independently reconstructed every earlier compiler proof or rerun the complete formal endpoint.

8.2 Reproduction and the scope of the checks

From the archive root, with a standard Python 3 installation, run

```
python3 reproduce.py
```

No network, computer algebra package, or proof assistant is required for the finite checks. To rebuild the manuscript, use a standard TEX installation:

```
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The arithmetic checks compare the compact norm evaluator with an independent eight-dimensional conjugate-product evaluation on signed assignments; exercise the inverse reconstruction including negative strong roots; verify transport shifts; audit live gates and port use; and check nonvanishing leading forms modulo two primes. The integer review separately derives the signed bounds, the two boundary exclusions, and the exact weighted leading form. The coefficient checks cover 69,230 basis convolution identities, all-rotation support facts, carry/slack inequalities, population counts, the five-adic unit, and the four exact payloads. The inherited language checker tests 425 explicitly halting control helices comprising 33,800 local tiles.

These tests detect transcription errors and verify the named finite properties. Finite fixtures do not establish a universal Diophantine equivalence. That conclusion depends on the proofs, including the inherited compiler, rather than on a search for zeros or a successful sample of computations. No new Lean/Rocq proof object for (8.1) is supplied.

8.3 Which presentation to use

For twelve positive unknowns, (5.14) is the recommended system: three equations, 110 operations, maximum degree 1072. For a single equation with that domain, use (5.2). For ordinary unrestricted integer unknowns, use (6.14) if a system is acceptable and (6.4) if one equation is required. The sum of squares is useful when its especially direct zero-set argument outweighs its higher degree.

The contribution improves the inspected moderate-degree twelve-positive and twenty-six-integer families. It does not dominate the retained 67-operation positive polynomial, the 59-operation larger-variable system, the degree-172 twenty-seven-integer product, or the nine-natural-variable Jones specialization. There is no proved global optimum among all Diophantine representations of this consistency statement.

A further variable-count comparison is timely. Sun's September 2026 preprint [12], Theorem 1.1, states that every recursively enumerable subset of $\mathbb{N}$ has a representation using ten integer existential unknowns. Its theorem gives a generic route to a ten-integer consistency instance. We have not instantiated or independently audited that compiler here, so this report makes no claim to a global integer-variable record.

8.4 Suggested next work

The best immediate verification target is to formalize the new signed separation and saturation composition as a small extension of the existing core interface. The positive proof's new obligation is similarly localized to (5.5)–(5.8). For further size reduction, the most promising open comparison is whether a weaker integer-valued or nonmonic inverse construction can remove another guarded coordinate without destroying soundness at zero or negative scales. Any such proposal must recheck coherent root recovery, the $C = 0$ boundary, exact leading cancellation, and simultaneous canonical completeness. The present work stops at the improvements proved and reproduced in this archive.

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